

# robocar-unified

## Single-Board AI Robot Car

Assembly, Wiring & Firmware Build Guide

A hands-on guide to assembling the consolidated robocar on a **Seeed Studio XIAO ESP32-S3 Sense**. Camera capture, Gemini AI planning, motor control, and peripherals all run on one module.

|            |                               |
|------------|-------------------------------|
| Target MCU | ESP32-S3 (8 MB PSRAM / flash) |
| Toolchain  | ESP-IDF v5.4 (containerized)  |
| Difficulty | Intermediate · ~2–3 h         |

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## 1 · Overview

The **robocar-unified** project consolidates the original dual-ESP32 robocar (a Heltec main controller plus an ESP32-CAM) onto a single **XIAO ESP32-S3 Sense**. One module now handles camera capture, AI inference, motor control, and all peripherals, using dual-core affinity to keep motor timing isolated from bursty network and vision work.

The control system is **hierarchical**: a slow **planner** (Core 1) calls Google’s Gemini Robotics-ER to emit structured goals, and a fast **reactive executor** (~30 Hz, Core 0) drives the robot smoothly toward those goals while an ultrasonic sensor provides an independent obstacle reflex. The planner is not on a fixed schedule: the board boots **dormant** and asks Gemini only when something happened, backing off 15 s → 300 s when nothing does.

### Core 0 — real-time

Reactive executor (visual servo, heading hold, motor PWM), peripheral I/O, command console, and the ultrasonic obstacle reflex.

### Core 1 — bursty I/O

Planner (Gemini calls), OV3660 camera DMA, audio, WiFi / MQTT / OTA.

### What you get

A two-wheel-drive car that captures frames, asks Gemini what to do, and drives toward goals — with pan/tilt camera, status LEDs, an OLED display, buzzer feedback, WiFi provisioning over the USB console, and over-the-air updates.

## 2 · Bill of Materials

| Qty | Component                                    | Notes                                               |
|-----|----------------------------------------------|-----------------------------------------------------|
| 1   | XIAO ESP32-S3 Sense                          | MCU + OV3660 camera + PDM mic + 8 MB PSRAM, USB-C   |
| 1   | TCA9548A I <sup>2</sup> C multiplexer        | Breakout, address 0x70                              |
| 1   | PCA9685 16-ch PWM driver                     | Breakout, address 0x40                              |
| 1   | TB6612FNG dual motor driver                  | Breakout                                            |
| 1   | SSD1306 OLED display                         | 128×64, I <sup>2</sup> C, address 0x3C              |
| 1   | MCP23017 GPIO expander                       | Breakout, address 0x20 — optional                   |
| 1   | MAX98357A I <sup>2</sup> S class-D amplifier | Mono, for voice output                              |
| 1   | Speaker                                      | 4–8 Ω, 2–3 W                                        |
| 1   | Ultrasonic rangefinder                       | <b>3.3 V variant:</b> HC-SR04P / RCWL-1601 / US-100 |
| 2   | RGB LED                                      | Common-anode                                        |
| 2   | SG90 micro servo                             | Pan / tilt                                          |
| 2   | DC gear motor + wheel                        | ~3–6 V hobby motors                                 |
| 1   | Piezo buzzer                                 | Passive                                             |
| 1   | 100 Ω resistor                               | In series with buzzer                               |
| 1   | Electrolytic capacitor                       | ≥470 μF — for MAX98357A supply                      |
| 2   | 18650 Li-ion cell + holder                   | Battery pack                                        |
| 1   | XL6009 boost converter                       | Regulated to 5 V                                    |
| —   | Chassis, wiring, headers                     | 2WD car chassis, jumper wires, standoffs            |

## Tools required

Soldering iron + solder, wire strippers, small screwdriver set, multimeter (for verifying 5 V rail and continuity), a USB-C cable, and a computer with Docker (for the containerized firmware build).

### Sensor voltage — read this

The ultrasonic sensor **must be a 3.3 V-compatible module** (HC-SR04P, not the classic 5 V HC-SR04). The XIAO's GPIOs are not 5 V-tolerant — a 5 V ECHO line can damage the board.

## 3 · System Architecture

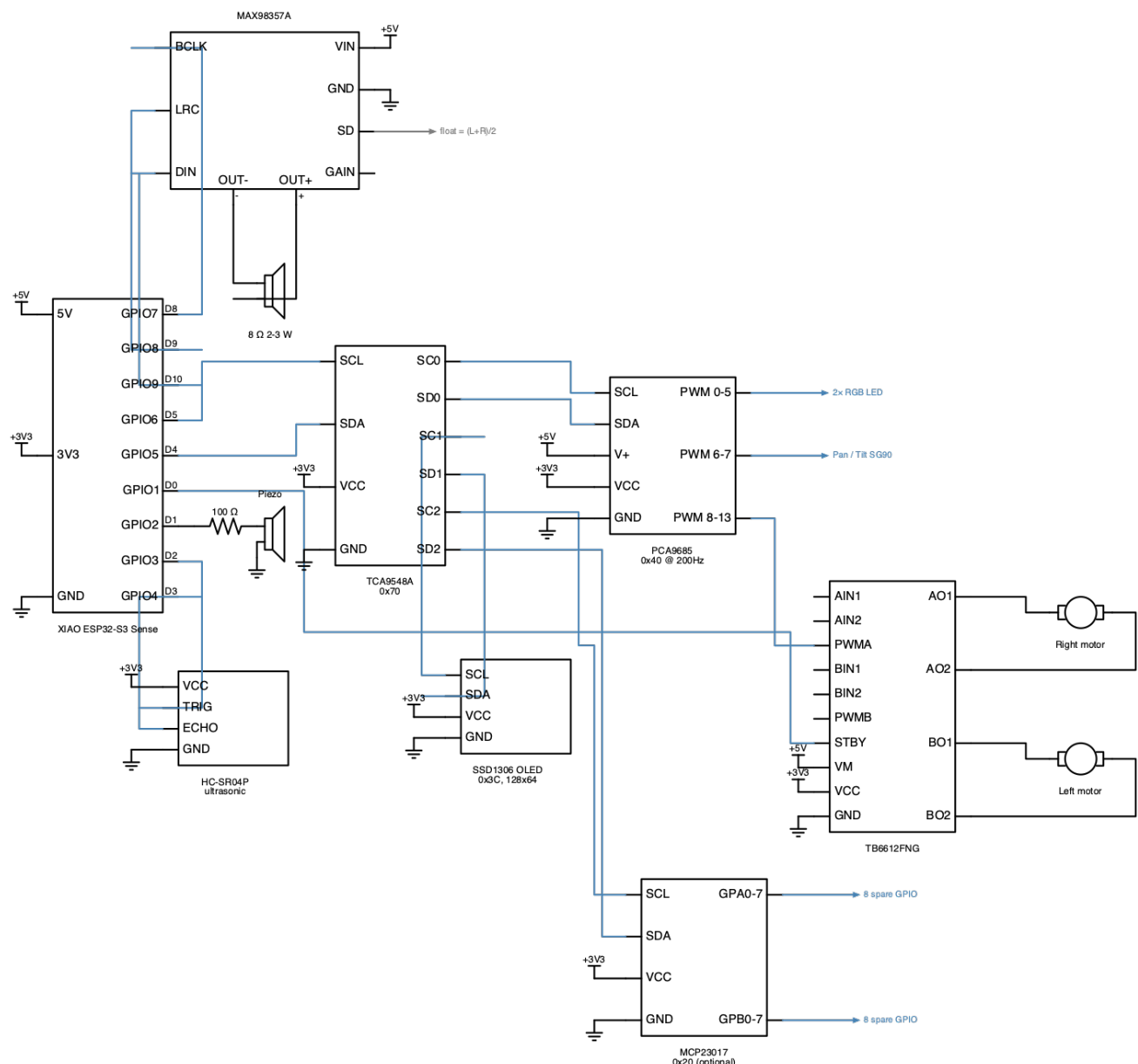


Figure 1: Full wiring schematic. Source: docs/schematics/circuits/robocar\_unified.py.

Everything on the I<sup>2</sup>C bus hangs off the **TCA9548A multiplexer** — the ESP32-S3 never talks to the PCA9685 or OLED directly. This lets devices that would otherwise share addresses coexist, and keeps the two 400 kHz devices on separate channels. The ESP32-S3 itself drives only five groups of signals directly: I<sup>2</sup>C (GPIO5/6), the motor-enable STBY line (GPIO1), the buzzer (GPIO2), the ultrasonic sensor (GPIO3/4), and **I<sup>2</sup>S audio** (GPIO7/8/9) to the MAX98357A amplifier.

## 4 · Wiring Reference

### 4.1 · XIAO ESP32-S3 GPIO assignments

Only 11 GPIOs are exposed on the XIAO headers. Camera pins are internal to the Sense module and do not conflict.

| XIAO Pin | GPIO   | Function                    | Notes                             |
|----------|--------|-----------------------------|-----------------------------------|
| D0       | GPIO1  | TB6612FNG STBY              | HIGH = motors enabled             |
| D1       | GPIO2  | Piezo buzzer                | LEDC PWM · 100 $\Omega$ in series |
| D2       | GPIO3  | Ultrasonic TRIG             | 10 $\mu$ s pulse output           |
| D3       | GPIO4  | Ultrasonic ECHO             | Pulse width in (RMT RX)           |
| D4       | GPIO5  | <b>I<sup>2</sup>C SDA</b>   | to TCA9548A                       |
| D5       | GPIO6  | <b>I<sup>2</sup>C SCL</b>   | to TCA9548A                       |
| D6       | GPIO43 | USB Serial TX               | Debug console                     |
| D7       | GPIO44 | USB Serial RX               | Debug console                     |
| D8       | GPIO7  | <b>I<sup>2</sup>S BCLK</b>  | to MAX98357A — bit clock          |
| D9       | GPIO8  | <b>I<sup>2</sup>S LRCLK</b> | to MAX98357A — word select        |
| D10      | GPIO9  | <b>I<sup>2</sup>S DIN</b>   | to MAX98357A — serial data        |

I<sup>2</sup>C runs at **400 kHz**.

#### GPIO budget fully allocated

There are no spare header pins. Additional digital I/O must go through the MCP23017 on TCA9548A channel 2.

### 4.2 · I<sup>2</sup>C topology (TCA9548A @ 0x70)

Select the channel on the multiplexer **before** addressing any downstream device. The MCP23017 on ch2 is optional — the firmware boots fine without it.

| Channel | Device                                       | Address       |
|---------|----------------------------------------------|---------------|
| ch0     | PCA9685 PWM driver (motors, servos, LEDs)    | 0x40 @ 200 Hz |
| ch1     | SSD1306 OLED display (128×64)                | 0x3C          |
| ch2     | MCP23017 GPIO expander (optional)            | 0x20          |
| ch3–7   | <i>reserved — IMU / ToF / future sensors</i> | —             |

### 4.3 · PCA9685 channel map (0x40, 200 Hz)

All motor direction, motor PWM, servo, and LED outputs go through the PCA9685. Motor direction pins use PCA9685 “full-on” (4096) / “full-off” (0); PWM pins use the full 12-bit range (0–4095).

| Ch | Signal           | Ch    | Signal              |
|----|------------------|-------|---------------------|
| 0  | Left LED — R     | 8     | Motor R — IN1 (dir) |
| 1  | Left LED — G     | 9     | Motor R — IN2 (dir) |
| 2  | Left LED — B     | 10    | Motor R — PWM       |
| 3  | Right LED — R    | 11    | Motor L — IN1 (dir) |
| 4  | Right LED — G    | 12    | Motor L — IN2 (dir) |
| 5  | Right LED — B    | 13    | Motor L — PWM       |
| 6  | Pan servo (SG90) | 14–15 | <i>reserved</i>     |

**Ch Signal****7 Tilt servo (SG90)**

200 Hz is a compromise between servo timing (ideal 50 Hz) and motor PWM smoothness — it works well for SG90s and the TB6612FNG.

**4.4 · Ultrasonic rangefinder**

| Signal | Pin        | Voltage | Function                              |
|--------|------------|---------|---------------------------------------|
| TRIG   | GPIO3 (D2) | 3.3 V   | 10 µs pulse triggers a measurement    |
| ECHO   | GPIO4 (D3) | 3.3 V   | Pulse width encodes distance (RMT RX) |
| VCC    | 3.3 V      | 3.3 V   | <b>3.3 V variant only</b>             |
| GND    | any GND    | —       | Shared ground                         |

The sensor samples at ~20 Hz. **Obstacle reflex:** if distance < 15 cm, the executor immediately stops and reverses, independent of planner goals.

**4.5 · Audio output (MAX98357A I<sup>2</sup>S amplifier)**

The robot speaks through a mono I<sup>2</sup>S class-D amplifier. Audio is 24 kHz 16-bit PCM — the native output rate of the Gemini TTS model — decoded incrementally into a PSRAM ring so playback starts before the download finishes.

| Signal  | Pin         | Function                                           |
|---------|-------------|----------------------------------------------------|
| BCLK    | GPIO7 (D8)  | Bit clock                                          |
| LRCLK   | GPIO8 (D9)  | Word select / left-right clock                     |
| DIN     | GPIO9 (D10) | Serial audio data to amplifier                     |
| Vin     | 5 V         | Power — see §5 for supply requirements             |
| GND     | any GND     | Shared ground                                      |
| SD_MODE | floating    | (L+R)/2 — firmware duplicates mono into both slots |
| GAIN    | float       | 9 dB default                                       |

The I<sup>2</sup>S channel is disabled between utterances — the MAX98357A hisses faintly whenever BCLK is running, so leaving it clocking silence is audible.

**Trade-off**

Wiring the amplifier on D8–D10 **replaces the microSD slot**. On the Sense expansion board these three pins are the SD card's SPI bus. There is no alternative pin set — I<sup>2</sup>S needs a hardware peripheral, so it cannot be moved behind the PCA9685 or the MCP23017.

**5 · Power**

Power the car from a **2×18650 pack** through an **XL6009 boost converter set to 5 V**. Distribute that 5 V rail to the XIAO 5 V pin, the TB6612FNG (VM + VCC), the PCA9685 (V+ and VCC), the servos, and the **MAX98357A amplifier (Vin)**.

The 3.3 V logic for the OLED, ultrasonic sensor, TCA9548A, and MCP23017 comes from the XIAO's 3V3 pin. Keep

**Golden rule**

**Common ground everywhere.** Every module — boost converter, XIAO, motor driver, PCA9685, servos, sensors — must share one GND. Missing grounds cause brown-outs, I<sup>2</sup>C lockups, and erratic motion.

motor/servo current (high, noisy) on the 5 V rail and logic on 3V3.

**Amplifier supply — read this:** The MAX98357A draws up to 1 A peaks into a 4  $\Omega$  load. With brown-out detection already disabled for motor inrush, an undersized rail will not warn you — it will present as random resets or corrupt audio mid-sentence. Fit a  $\geq 470\text{ }\mu\text{F}$  **bulk capacitor** at the amplifier's Vin, plus the usual 0.1  $\mu\text{F}$  close to the pin. Prefer a **separate 5 V feed** from the boost converter to the amplifier rather than daisy-chaining off the motor-driver rail. An **8  $\Omega$  speaker** roughly halves peak current versus 4  $\Omega$ .

Brown-out detection is disabled in firmware because motor inrush was tripping it; a stiff 5 V supply and thick power wires matter.

## 6 · Assembly Steps

1. **Mount the mechanics.** Fit the two gear motors and wheels to the chassis, add the caster/third wheel, and mount the battery holder low and centered.
2. **Wire power first.** Connect the 18650 pack to the XL6009 input, set its output to **5.0 V with a multimeter before connecting anything else**, then run the 5 V and shared GND rails.
3. **Place the XIAO** and bring out I<sup>2</sup>C (GPIO5/6), STBY (GPIO1), buzzer (GPIO2), ultrasonic pins (GPIO3/4), and I<sup>2</sup>S (GPIO7/8/9).
4. **Wire the I<sup>2</sup>C chain:** XIAO SDA/SCL → TCA9548A → PCA9685 (ch0), OLED (ch1), and optionally MCP23017 (ch2). Pull-ups on the breakouts are usually sufficient.
5. **Wire the motor driver:** PCA9685 ch8–13 → TB6612FNG inputs; STBY → GPIO1 motor outputs → the two DC motors; VM/VCC → 5 V.
6. **Add servos** (PCA9685 ch6/7) and **RGB LEDs** (ch0–5, common-anode).
7. **Add the buzzer** on GPIO2 through the 100  $\Omega$  resistor, and the ultrasonic sensor on GPIO3/4 (3.3 V power).
8. **Wire the audio path:** MAX98357A BCLK/LRC/DIN → GPIO7/8/9, Vin → 5 V with  $\geq 470\text{ }\mu\text{F}$  bulk cap, speaker → amp output. Leave SD\_MODE floating for (L+R)/2.
9. **Double-check the 3.3 V vs 5 V rails** and confirm common ground with a multimeter continuity test before first power-up.

## 7 · Build & Flash the Firmware

Builds run inside a container — **no local ESP-IDF install is required**. The XIAO uses native USB-Serial-JTAG (USB VID 0x303a), which just auto-detects.

```
# From the repo root
just robocar-unified::build                # containerized ESP-IDF v5.4 build
PORT=/dev/cu.usbmodem* just robocar-unified::flash # flash over USB-C
just robocar-unified::monitor              # serial console
# or, in one step:
just robocar-unified::flash-monitor
```

### Can't enter download mode?

Hold **BOOT**, tap **RESET**, then release BOOT to force the bootloader, and re-run the flash command.

The flasher writes three images to an 8 MB, OTA-capable layout:

| Offset | Image                                     | Partition       |
|--------|-------------------------------------------|-----------------|
| 0x0    | build/bootloader/bootloader.bin           | bootloader      |
| 0x8000 | build/partition_table/partition-table.bin | partition table |

| Offset  | Image                     | Partition   |
|---------|---------------------------|-------------|
| 0x12000 | build/robocar-unified.bin | ota_0 (app) |

## 8 · First Boot & Provisioning

### 8.1 · WiFi over the USB console (Improv Serial)

No WiFi credentials are compiled in. On first boot the device speaks **Improv Serial** on the same USB port you flashed it with – not the BLE variant, so there is nothing to pair. Open the board in a browser-based Improv Serial provisioner (Chrome desktop, e.g. ESP Web Tools) and send your SSID and password. They are written to NVS only after they are proven to connect, so a typo cannot overwrite a working network.

For local development you can instead copy `main/credentials.h.example` to `main/credentials.h` (gitignored) and hard-code credentials.

### 8.2 · Discovery & AI backend

After connecting, the car is reachable at **robocar-unified.local** via mDNS. The planner uses **Gemini Robotics-ER 2** to emit goals — `drive()`, `track()`, `rotate()`, and `stop()` — plus `speak()`, which renders a sentence through a second TTS model and plays it while the robot keeps driving.

### 8.3 · Over-the-air updates

OTA is enabled with app rollback. The updater pulls releases from the `laurigates/mcu-tinkering-lab` GitHub repo; `version.txt` is the single source of truth for the running version and is managed by `release-please`.

## 9 · Functional Checkout

Work through these after first flash, watching the serial monitor:

| ✓                        | Check                 | Expected result                                                               |
|--------------------------|-----------------------|-------------------------------------------------------------------------------|
| <input type="checkbox"/> | Boot log              | No PSRAM / boot-loop errors; tasks pin to cores 0 & 1                         |
| <input type="checkbox"/> | I <sup>2</sup> C scan | TCA9548A, PCA9685, and OLED all detected                                      |
| <input type="checkbox"/> | OLED                  | Status screen renders (128×64)                                                |
| <input type="checkbox"/> | LEDs                  | Both RGB LEDs cycle / show status colors                                      |
| <input type="checkbox"/> | Buzzer                | Audible startup tone                                                          |
| <input type="checkbox"/> | Servos                | Pan/tilt center, then sweep within limits                                     |
| <input type="checkbox"/> | Motors                | Both wheels drive forward and reverse; STBY HIGH                              |
| <input type="checkbox"/> | Ultrasonic            | Distance readings track a hand moving closer/away                             |
| <input type="checkbox"/> | Reflex                | Car stops/reverses when an obstacle is < 15 cm                                |
| <input type="checkbox"/> | Audio / TTS           | Robot speaks startup message; speech queues without blocking motion           |
| <input type="checkbox"/> | Microphone            | mic reports frames arriving, not gate: DEAF; listen answers a spoken question |
| <input type="checkbox"/> | WiFi                  | Provisions via Improv; <code>robocar-unified.local</code> resolves            |



## 10 · Troubleshooting

| Symptom                           | Likely cause & fix                                                                                                             |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Boot loop on power-up             | Wrong PSRAM mode. The Sense uses <b>octal</b> PSRAM ( <code>CONFIG_SPIRAM_MODE_OCT=y</code> ); don't change it.                |
| Random resets under motor load    | Weak 5 V rail / missing common ground. Use thicker power wires and verify the XL6009 holds 5 V under load.                     |
| No I <sup>2</sup> C devices found | Not selecting the TCA9548A channel first, or SDA/SCL swapped. Check <code>GPIO5=SDA</code> , <code>GPIO6=SCL</code> .          |
| OLED and PCA9685 conflict         | Both bypassing the mux. Route each through its own TCA9548A channel (ch1 / ch0).                                               |
| Motors don't move                 | STBY (GPIO1) not HIGH, or VM not on 5 V. Confirm TB6612FNG power and enable line.                                              |
| Servos jitter                     | Shared noisy rail. Keep servo power on 5 V with common ground; 200 Hz PWM is expected.                                         |
| Board won't flash                 | Force download mode: hold BOOT, tap RESET, release BOOT.                                                                       |
| Damaged ECHO / no distance        | Used a 5 V HC-SR04. Replace with a 3.3 V module (HC-SR04P).                                                                    |
| No audio / distorted speech       | Weak MAX98357A supply. Fit $\geq 470\ \mu\text{F}$ bulk cap at Vin, or use a separate 5 V feed. Check I2S wiring on GPIO7 – 9. |